OCCUPATIONAL HEALTH, SAFETY & ENVIRONMENT

QADIRPUR GAS PLANT

LIVE PLANT OPERATIONS / MAINTENANCE INTERFACE RISK ASSESSMENT

CONTROLLED DOCUMENT | ACTIVITY-BASED HAZARD AND RISK REGISTER | PUBLIC-SOURCE PRE-ASSESSMENT

DOCUMENT STATUS: NOT APPROVED FOR EXECUTION — SITE DATA, WORK PACK AND OWNER REQUIREMENTS MUST BE VERIFIED

Public-source pre-assessment — site validation and authorized approval required. This document follows a generic professional reference format for layout guidance; it is NOT an official OGDCL form, is NOT OGDCL-approved, and is not evidence of a site inspection. Public information on Qadirpur is limited and largely historical — see Sections 3 and 14.

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1. DOCUMENT CONTROL & PROJECT INFORMATION

Complete every applicable field with verified information. Use "Not applicable" only where justified; do not invent missing site information.

Project / Facility OGDCL Qadirpur Gas Processing Plant — Qadirpur Gas Field/Plant, Ghotki, Sindh, Pakistan [S-01][S-03]	**Client / Asset Owner** Oil & Gas Development Company Limited (OGDCL) — operator [S-01][S-05]. Joint venture as reported in 2008: OGDCL 75%, PKP Exploration 9.5%, KUFPEC Pakistan 8.5%, PPL 7% [S-03] — current JV shares [CONFIRM]	**Principal Contractor** [SITE INPUT REQUIRED — contractor(s) for the selected work package are not publicly confirmed]	**Subcontractor(s)** [SITE INPUT REQUIRED — not publicly disclosed]
Assessment Title / Activity Live plant operations / maintenance interface risk assessment — maintenance, modification, tie-in and commissioning-type activities interfacing with the operating gas plant, where confirmed in the approved work pack	**Work Location / Area** Qadirpur Gas Plant, near Ghotki, Sindh [CONFIRM exact unit / plot with the approved plot plan]	**RA Number / Document ID** QAD-RA-001 [TO BE CONFIRMED BY DOCUMENT CONTROL]	**Revision / Status** Rev. 0 — Public-source pre-assessment — NOT APPROVED FOR EXECUTION
Assessment Date 28 October 2026 (public-source research snapshot)	**Review / Expiry Date** [PER OGDCL PROCEDURE — to be set at approval; review triggers are listed in Section 15]	**Related PTW / JSA / Method Statement** [SITE INPUT REQUIRED — verified references to be entered from the site PTW register and the approved work pack]	**Applicable Procedure / Standard** OGDCL HSE / PTW procedures [NOT REVIEWED — controlled owner documents, S-01]; Sindh OSH Act 2017 + Rules 2019 (framework, summarised) [S-07]; ISO 45001:2018 + Amd 1:2024 (framework) [S-08]
Prepared By / Role [REAL NAME / TITLE / SIGNATURE — to be completed by an authorized competent person]	**Reviewed By / Role** [REAL NAMES / TITLES — Operations, Maintenance, Process/Mechanical/Electrical, HSE]	**Approver / Role** [AUTHORIZED APPROVER ONLY — this Rev. 0 pre-assessment is NOT approved]	**Workforce / Shifts** Current site headcount NOT PUBLICLY DISCLOSED — obtain gate / access-control and contractor mobilisation records (Section 5)

1.1 Document control detail

Document control item	Entry
Document number	QAD-RA-001 [TO BE CONFIRMED BY DOCUMENT CONTROL]
Project / contract number	[CONFIRM — to be assigned from the actual work package]
Facility / area	Qadirpur Gas Plant, Ghotki District, Sindh, Pakistan — exact plot/area [CONFIRM WITH APPROVED PLOT PLAN]
Assessment boundary	The operating Qadirpur gas processing plant and maintenance/construction interfaces at that single location only. EXCLUDES: Qadirpur well pads and gathering network beyond the plant boundary, other OGDCL fields/plants, and any separate power-plant facilities supplied from the field — each requires its own assessment.
Work package / work order	[CONFIRM — assessment applies only to activities confirmed in the approved work pack]
Revision	Rev. 0 — Public-source pre-assessment
Assessment (issue) date	28 October 2026 (public-source research snapshot)

Document control item	Entry
Date of site walkdown / work-pack review	[TO BE COMPLETED — no site visit has been made for this pre-assessment]
Prepared by	[REAL NAME / TITLE / SIGNATURE — to be completed by an authorized competent person]
Reviewed by	[REAL NAMES / TITLES / SIGNATURES — Operations, Maintenance, Process/Mechanical/Electrical, HSE]
Approved by	[REAL NAME / TITLE / SIGNATURE — authorized approver only. This document is NOT approved.]
Classification	Working pre-assessment — not an official OGDCL document; not OGDCL-approved; contains no OGDCL branding and claims no review of controlled OGDCL manuals.

VALIDATION DISCLAIMER: This assessment was prepared from publicly available sources only (see Section 17 references, access date 28 October 2026). No site visit, worker interview, permit review, drawing review or headcount verification has taken place. Every site-specific value that could not be verified from public sources is marked [CONFIRM] / [SITE INPUT REQUIRED] and must be established from controlled site documents and a competent site walkdown before this document is approved or used to authorize work.

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2. Executive summary

Why this assessment exists. Qadirpur is an operating gas field and processing plant in Ghotki District, Sindh, operated by OGDCL [S-01][S-05]. The field was discovered in 1990 and regular production commenced in September 1995 after gas gathering and processing facilities were installed [S-03]; the plant's publicly documented core process is membrane separation / purification of natural gas, with compression added as reservoir pressure declined [S-02][S-04]. Maintenance, modification and tie-in work therefore interfaces with a live, pressurised hydrocarbon plant. That live-plant interface is the highest-risk feature addressed by this assessment.

Top major-accident hazards identified (Section 7): loss of containment of flammable gas/condensate with jet fire, flash fire or vapour cloud explosion (RA-02); line opening on incompletely isolated or unproven systems (RA-05); confined-space entry (RA-06); possible toxic-gas (H2S) exposure, which is a VERIFICATION-CRITICAL hazard because current gas composition is not public (RA-03); failure of isolation/LOTO (RA-04); heavy lifting over or near live plant (RA-12); and breakdown of the emergency-response interface for transient maintenance/construction personnel (RA-21). Heat stress in the Sindh summer is a critical occupational-health hazard (RA-16).

Critical controls and hold points. Work must not start until, as a minimum: the SIMOPS interface is reviewed and authorised by operations; isolations are proven with zero-energy verification; current gas composition data has been received and the H2S decision recorded; the OGDCL risk matrix has replaced the proposed matrix in Section 6; the approved emergency plan details are briefed to every worker; and actual shift headcount is being captured for muster reconciliation. Each hold point is stated on the relevant register row and in Section 8.

Unresolved inputs — and a warning on the public record. The public record for Qadirpur is THIN and largely historical: the most detailed capacity and production figures found are from 2004–2016, and no current headcount, workfront list, process conditions or equipment list is publicly disclosed. Nothing of that kind is estimated anywhere in this document. Section 14 lists every site input required before competent review and approval. Residual risk is therefore NOT declared acceptable by this pre-assessment: residual ratings are provisional professional judgements that only become valid when the stated controls are implemented, verified and accepted by the authorized owner representative.

3. Project profile and source-backed facts

Labels used throughout: 'Verified public fact' = checked against the cited source on 28 October 2026; 'Historically reported' = true for its stated period/scope only; 'Technical consideration — not yet confirmed at site' = a credible hazard or control topic that site data must confirm or rule out; 'Site input required' = information that only controlled site documents can supply.

Fact	Value / statement	Status & source	What it does — and does not — prove
Operator / facility	Oil & Gas Development Company Limited (OGDCL) is the operator; the facility is officially named 'Qadirpur Gas Field/Plant, Ghotki Sindh' in OGDCL tender documents.	Verified public fact [S-01][S-05]	Establishes the real plant and operator. Does NOT establish the assessment's exact plot boundary — plot plan required.
Location	Qadirpur, Ghotki District, Sindh, Pakistan — about 8 km from Ghotki; also described as 70 km NE of Sukkur and 100 km E of Jacobabad.	Verified public fact [S-03]; secondary detail [S-04]	Establishes geography and the Sindh (Pakistan) legal context. Exact plant coordinates/plot are NOT used — obtain from the approved plot plan.
Discovery / start-up	Field discovered in 1990 (March 1990 in secondary sources); regular production commenced September 1995 after installation of gas gathering facilities and processing plant; developed in three phases until March 2004.	Verified public fact [S-03][S-04]	Proves a long-operating, mature plant — ageing-asset and declining-pressure considerations apply. Does NOT describe current plant condition.
Operating status (current)	Global Energy Monitor lists Qadirpur as an OPERATING oil and gas field with OGDCL as operator (page published/updated 27 Oct 2026); OGDCL's Annual Report 2024 lists Qadirpur among its major producing gas fields.	Verified public fact [S-05][S-06]	Supports treating Qadirpur as a live operating plant in 2026. Does NOT give today's production rate, headcount or workfronts.
Core process (publicly documented)	Membrane separation/purification of natural gas: the membrane process removes water vapour and carbon dioxide [S-04]; the 2004 expanded plant used UOP cellulose acetate plus membrane technology described as for CO2 and H2S removal and hydrocarbon recovery [S-03]; pretreatment in 2004 included a chiller, a MemGuard unit and a Separex membrane system [S-02].	Verified public fact — as described in 2004–2017 sources [S-02][S-03][S-04]	Supports the gas-processing/dehydration-type hazard profile in Section 7 at a general level ONLY. It does NOT confirm the current plant configuration, current gas composition, or that any named 2004 equipment is still in service — all [CONFIRM] from current P&IDs.
Compression (historical)	Compression facilities were being installed to address declining reservoir pressures and maintain sales-gas delivery pressure; a 600 MMscfd processing facility was installed in 2006–07.	Historically reported [S-04]	Supports including compression-system interfaces in scope where the work pack confirms them. Historical only — current compressor inventory is [SITE INPUT REQUIRED].
Capacity figures (historical ONLY)	235 MMscfd membrane capacity expanded to 400 MMscfd sales gas in 2004, with a further rise to 500 MMscfd then planned [S-02]; compression project reported in 2011-era sources aimed to maintain a 550–600 MMcfd plateau (see Section 5 table).	Historically reported — 2004 / 2011 scope and dates [S-02][S-04]	Scale context ONLY. These figures must NOT be used as current production, current capacity or design data for any work.

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Fact	Value / statement	Status & source	What it does — and does not — prove
Joint venture (as reported 2008)	OGDCL 75% (operator), PKP Exploration 9.5%, KUFPEC Pakistan 8.5%, Pakistan Petroleum Ltd (PPL) 7%.	Historically reported [S-03]	Ownership context only. Current JV shares and any partner HSE requirements must be confirmed — they do not change day-to-day permit authority, which is a site input.
Owner HSE requirements exist	An OGDCL tender for Qadirpur states that the contractor and its labour will follow OGDCL guidelines/policies related to HSE, and that contractor procedures follow ISO/API codes for the work described.	Verified public fact [S-01]	Proves owner HSE requirements bind contractors at this plant — and that the controlled texts were NOT reviewed for this pre-assessment. They must be obtained through authorized channels.
Legal framework — Sindh, Pakistan	The Sindh Occupational Safety and Health Act, 2017 (Sindh Act No. I of 2018) is the main provincial OSH framework in Sindh, with detailed requirements in the Sindh OSH Rules, 2019 (as summarised by a secondary source).	Verified public fact — PAKISTAN/SINDH LAW as summarised [S-07]	Sets the correct jurisdiction for this assessment: Pakistan/Sindh, NOT Saudi Arabia. Exact official texts and plant-specific petroleum/environmental requirements must be confirmed in the project legal register (Section 6.2 / Section 14).
Management-system framework	ISO 45001:2018, confirmed in 2024, remains current; Amendment 1:2024 (climate action) applies.	Verified public fact [S-08]	Provides the OH&S management framework. It is NOT a hazard checklist and NOT Pakistan law.

4. Site footprint, asset and workfront register

Public sources describe the facility elements below only at a general level. No public source provides a current approved workfront register, package split or plot boundary; every operational field is therefore [SITE INPUT REQUIRED]. Other OGDCL fields are listed only to record their exclusion.

Facility / asset / workfront	Location	Package / contractor	Current activity & SIMOPS interface	Operational status	Source / drawing reference	Data owner	Verification status
Qadirpur gas processing plant — operating process areas (general)	Ghotki District, Sindh [CONFIRM plot/unit IDs]	[CONFIRM package]	Live operation; maintenance/modification interface where tie-ins occur	Operating [S-05][S-06]	Approved plot plan [SITE INPUT REQUIRED]	Operations	Not verified — site input required
Membrane separation / purification area (process as publicly described)	Qadirpur plant [CONFIRM]	[CONFIRM]	Maintenance interface only where the work pack confirms it	Process described in 2004–2017 sources [S-02][S-03][S-04]; current configuration [CONFIRM]	Historical sources; current P&IDs [SITE INPUT REQUIRED]	Operations / Process	Process type verified historically; current configuration NOT verified
Gas compression area (process as publicly described)	Qadirpur plant/field [CONFIRM]	[CONFIRM]	Maintenance interface only where the work pack confirms it	Compression added historically for declining pressure [S-04]; current status [CONFIRM]	Historical source [S-04]; equipment list [SITE INPUT REQUIRED]	Operations / Mechanical	Not verified — do not assume any compressor type, count or duty
Gas gathering / well interfaces	Qadirpur field area [CONFIRM]	[CONFIRM]	Interface only where the approved work pack confirms it	Gathering facilities installed before 1995 start-up [S-03]; current status [CONFIRM]	Historical source [S-03]	Operations	EXCLUDED from the plant boundary unless the work pack adds it — drilling/workover not in scope
Utilities, condensate handling and export/dispatch interfaces	Qadirpur plant [CONFIRM]	[CONFIRM]	Maintenance interface only where the work pack confirms it	[CONFIRM from current plant documentation]	[SITE INPUT REQUIRED]	Operations	Not verified — no inventory, tank or export detail is assumed
Fabrication / laydown / contractor work areas	[CONFIRM locations]	[CONFIRM]	Prefabrication, storage and handling; traffic interface. OGDCL tender practice places shifting/lifting and housekeeping duties on contractors for specific jobs [S-01]	[CONFIRM current arrangement]	Tender example [S-01]; site plan [SITE INPUT REQUIRED]	Maintenance / Construction Manager	Not verified — confirm current arrangement
Other OGDCL fields/plants (e.g. Uch, Dakhni, KPD-TAY)	Other districts/provinces — SEPARATE SITES	n/a	NONE — excluded from this assessment	Operating fields in their own right [S-06]	Annual Report 2024 [S-06]	n/a	EXCLUDED — separate assessments required
Power facilities supplied from the field (third-party plants near Ghotki/Daharki as reported historically)	Separate facilities — NOT the Qadirpur plant	n/a	NONE — excluded; gas-supply interface only if the work pack confirms it	As reported historically [S-04]	Secondary source [S-04]	n/a	EXCLUDED — separate assessment required if work is performed there

5. Workforce and exposure register

5.1 Public figures — with date, scope and correct use

Figure	Value	Date / scope	Correct use in this assessment
Total persons physically inside the Qadirpur plant boundary on the issue date	NOT PUBLICLY DISCLOSED	Current — not found in any public source reviewed on 28 Oct 2026	MUST be obtained from the gate/access-control report and contractor mobilisation / daily manpower reports. Do not estimate.

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Figure	Value	Date / scope	Correct use in this assessment
OGDCL company-wide production (context only)	Gas 717 MMcf/day; crude oil 33,117 bbl/day (company totals)	OGDCL Annual Report 2024 [S-06]	COMPANY context ONLY. These are OGDCL totals across all fields — NOT Qadirpur production and NOT a workforce figure.
Historical Qadirpur figures (context only)	e.g. membrane expansion to 400 MMscfd sales gas (2004) [S-02]; 600 MMscfd processing facility installed 2006-07 [S-04]; FY2015-16 net production as reported [S-04]	2004 / 2006-07 / FY2015-16 — each labelled by its own date	HISTORICAL context ONLY, showing the plant's scale and decline/compression history. None of these is a current rate, a current capacity for design purposes, or a headcount.
Number of active workfronts / packages on the issue date	NOT CONFIRMED IN PUBLIC SOURCES	Current	The live workfront register must come from site management and the approved work pack.

WARNING: No public figure found in this research is a current Qadirpur headcount. OGDCL company totals and historical Qadirpur production/capacity figures measure different things, in different years, and none of them may be used for muster, welfare, transport or emergency planning.

5.2 Actual manpower and exposure register — TO BE COMPLETED FROM SITE RECORDS

Source of count required: gate access-control / badge report, muster roster, contractor mobilisation register or daily manpower report. Until a dated record is attached, every field below remains 'Not publicly disclosed — obtain actual site record.'

Field	Value (from site record only)	Record reference
Headcount date and exact time of count	Not publicly disclosed — obtain actual site record	
Shift pattern in force (day / night / both) [CONFIRM]	Not publicly disclosed — obtain actual site record	
Direct operator (OGDCL) personnel inside boundary — day shift	Not publicly disclosed — obtain actual site record	
Direct operator personnel — night shift	Not publicly disclosed — obtain actual site record	
Maintenance / main contractor personnel — day shift	Not publicly disclosed — obtain actual site record	
Maintenance / main contractor personnel — night shift	Not publicly disclosed — obtain actual site record	
Subcontractor personnel (list each subcontractor separately) — day	Not publicly disclosed — obtain actual site record	
Subcontractor personnel — night	Not publicly disclosed — obtain actual site record	
Vendor / OEM / specialist personnel	Not publicly disclosed — obtain actual site record	
Visitors inside boundary	Not publicly disclosed — obtain actual site record	
TOTAL persons inside the defined boundary (this count)	Not publicly disclosed — obtain actual site record	
Peak concurrent workforce in the reporting period	Not publicly disclosed — obtain actual site record	
Personnel working at height (this shift)	Not publicly disclosed — obtain actual site record	
Confined-space entrants and attendants (this shift)	Not publicly disclosed — obtain actual site record	
Lifting crew (operators, riggers, signallers)	Not publicly disclosed — obtain actual site record	
Hot-work crew and fire watches	Not publicly disclosed — obtain actual site record	
Electrical / instrument crew	Not publicly disclosed — obtain actual site record	
Plant / traffic interface personnel	Not publicly disclosed — obtain actual site record	
Emergency-response personnel on shift (fire, medical, rescue)	Not publicly disclosed — obtain actual site record	
Prepared by (name/role) / Verified by (name/role) / Revision & reporting period	Not publicly disclosed — obtain actual site record	

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6. Assessment methodology and applicable requirements

Method. Work is decomposed into activity steps (Section 7). For each step the assessor identifies the hazard source and the specific hazardous event, the persons exposed and how, and the credible consequences — including fatality and major-accident consequences where they are foreseeable. Controls already in place are recorded separately from additional controls, and each existing control is marked with its evidence status: in this pre-assessment almost all are 'to be verified', because no site evidence has been reviewed. Additional controls follow the hierarchy of controls (elimination, substitution, engineering, administrative, PPE), prioritising isolation and engineering barriers over procedures and PPE. A risk rating is never reduced merely by writing 'follow procedure' or adding PPE.

Three different risk values. (1) Initial/inherent risk assumes no controls beyond those physically inherent in the task. (2) Current risk credits only existing controls that are implemented AND verified by evidence. (3) Residual risk is rated only after the additional controls are implemented and independently/competently checked. In this draft, initial and residual ratings are PROVISIONAL PROFESSIONAL JUDGEMENTS made for comparison and prioritisation; they must be re-rated in the site validation workshop.

Major-accident limitation. A 5x5 matrix is a screening and prioritisation tool. It is NOT sufficient on its own for major-accident process hazards: a low likelihood score must never be used to dismiss a catastrophic scenario (RA-02, RA-03). Such scenarios require the project's specialist studies (HAZID, HAZOP, LOPA, QRA, bow-tie) and the interface to those studies is a site input (Section 14). This document is an activity-based HSE risk assessment; it does not replace a process hazard analysis, a method statement, or task-level JSAs (Section 9).

6.1 Proposed 5 x 5 risk matrix — PROPOSED WORKING MATRIX ONLY

Use the OGDCL/client-approved risk matrix whenever one exists. The 5 x 5 matrix below is a PROPOSED reference method only and must be replaced or formally accepted by the responsible organization before use. Risk score = Likelihood x Severity. The owner risk matrix was not available; likelihood/severity definitions are proposed qualitative definitions pending approval.

PROPOSED 5 x 5 RISK MATRIX

S x L	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

RATING DEFINITIONS (TO BE APPROVED)

LIKELIHOOD SCALE

Level	Term	Definition / guide
1	Rare	Exceptional; highly unlikely in the defined conditions.
2	Unlikely	Not expected, but credible.
3	Possible	Could occur under some circumstances.
4	Likely	Expected to occur in many circumstances.
5	Almost certain	Expected to occur frequently / repeatedly.

SEVERITY SCALE

Level	Term	Definition / guide
1	Insignificant	No injury or negligible harm / damage.
2	Minor	First-aid case or minor, reversible damage.
3	Moderate	Medical-treatment / lost-time potential; significant reversible harm or damage.
4	Major	Single-fatality potential, life-changing injury or major damage.
5	Catastrophic	Multiple fatalities or catastrophic asset / environmental consequences.

PROPOSED SCORING BANDS (subject to project approval): 1–4 Low — manage by routine controls. 5–9 Moderate — specific controls and supervision. 10–16 High — additional controls, documented review and authorized acceptance required before work. 17–25 Extreme — work must not proceed until risk is reduced and acceptance is given at the level the owner's system requires. These bands are proposed working bands, not a universal ISO/OGDCL matrix. High/Extreme residual risks in Section 7 carry explicit hold points.

6.2 Applicable requirements and verification register

Requirement / source	Jurisdiction & type	Relevant topic (summarised, not reproduced)	Applicability	Version/date checked	Owner/client document still needed	Verification status
Sindh Occupational Safety and Health Act, 2017 + Sindh OSH Rules, 2019 [S-07]	Pakistan / Sindh — LAW / REGULATORY	General employer duties for safety and health at workplaces in Sindh; safety/health arrangements; representatives and committees; notification and investigation of accidents, dangerous occurrences and occupational illnesses (as	All work at this Sindh plant	Act: Sindh Act No. I of 2018; Rules 2019; summary checked 28 October 2026	Official gazette texts; project legal register confirming applicability and any plant-specific duties	Framework verified from a secondary summary; official text outstanding

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Requirement / source	Jurisdiction & type	Relevant topic (summarised, not reproduced)	Applicability	Version/date checked	Owner/client document still needed	Verification status
		summarised — official text governs)				— SITE INPUT REQUIRED
OGDCL HSE guidelines / policies and plant PTW system [S-01]	Owner-specific — CONTRACTUAL / OPERATIONAL	Contractors at Qadirpur must follow OGDCL HSE guidelines/policies; work procedures per applicable ISO/API codes for the scopes described	All work — governing documents for execution	Tender statement checked 28 October 2026; controlled HSE/PTW texts NOT reviewed	Current authorized copies via the plant's document control	NOT REVIEWED — critical site input
Pakistan petroleum / gas and environmental requirements applicable to the plant (regulator and exact instruments to be confirmed)	Pakistan — LAW / REGULATORY	Upstream petroleum, gas transmission/dispatch and environmental protection duties that may apply to a gas processing plant	Plant operation and any modification work	Not verified in this check — do not cite instruments or clause numbers from memory	Project legal register prepared by the HSE/legal function	SITE INPUT REQUIRED
ISO 45001:2018 + Amd 1:2024 [S-08]	International — MANAGEMENT SYSTEM STANDARD	Framework for hazard identification, worker participation, operational control, emergency preparedness and review	Assessment process framework	Confirmed current 2024; checked 28 October 2026	Plant HSE management plan mapping	Verified as framework only — not a legal requirement
OSHA PSM overview; confined spaces; welding guidance [S-09][S-10][S-11]	United States — TECHNICAL GUIDANCE ONLY	Process-safety, confined-space and hot-work control concepts	Technical reference for RA-01/02/03/06/07/08	As cited in the research dossier	OGDCL process-safety, confined-space and hot-work procedures (governing)	Guidance only — NOT Pakistan law

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7. Detailed activity-based risk assessment register

Use one entry for each meaningful work step / hazard scenario. Hazards are described specifically; existing controls are distinguished from additional controls; residual risk is rated only for controls that are implemented and verified. 22 specific entries, tied to gas-processing work at a live plant. Ratings use the PROPOSED matrix in Section 6.1 and are professional judgements for prioritisation only — they must be re-rated in the site validation workshop with the OGDCL matrix. 'Existing controls' in this pre-assessment are stated with their evidence status; a control without evidence is NOT credited in any residual rating until verified on site.

RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
RA-01	Maintenance / construction work adjacent to operating gas-processing equipment (SIMOPS interface)	Qadirpur Gas Plant — live process areas [CONFIRM unit/plot]	Impact, dropped load, vehicle strike or hot particle contact with live hydrocarbon equipment causing loss of containment (LOC). Conflicting permits in one area.	Hydrocarbon release, fire/explosion, multiple fatalities, major plant damage and escalation. Persons at risk: Maintenance/construction crews, plant operators and other contractors in the affected area; exposure by proximity to live equipment.	TO BE VERIFIED: OGDCL tender documents state contractors must follow OGDCL HSE guidelines/policies [S-01], but no site PTW/SIMOPS evidence was reviewed for this pre-assessment.	3 x 5 = 15 — HIGH	ELIMINATION: schedule intrusive/adjacent work during a planned shutdown/turnaround where feasible. ENGINEERING: physical barriers and defined equipment-protection zones on the plot plan. ADMIN: SIMOPS review and written operations authorisation before work; daily coordination with the Area Authority; live permit map for the area; suspension rule on any plant upset. PPE: task PPE per site standard (last line only).	Area Authority / Operations Representative Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	OGDCL PTW (as defined by site system); SIMOPS authorisation; approved plot plan extract Review trigger / reference: Review on any change of plant status or workfront. Ref: OSHA PSM overview (technical guidance only) [S-09].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no work within the agreed protection zone of live equipment without signed operations authorisation. Stop work on any plant alarm/upset until cleared by the Area Authority.
RA-02	Work on or near pressurised hydrocarbon gas / condensate systems	Processing, membrane, compression and export systems in package [CONFIRM]	Uncontrolled release of flammable gas/condensate forming a flammable cloud; jet fire, flash fire or vapour cloud explosion with escalation.	Multiple fatalities, major asset loss, environmental release; credible major-accident event. Persons at risk: Crews at the release point, downwind personnel, plant operators; exposure via thermal radiation, blast and projectiles.	TO BE VERIFIED: fixed gas detection, ESD and fire-protection status at the interface not evidenced — site input required. A 5x5 matrix alone is not sufficient for this scenario; HAZID/HAZOP/bow-tie interface required (Section 6).	3 x 5 = 15 — HIGH	ELIMINATION: defer hydrocarbon introduction until work in that system is complete. ENGINEERING: positive isolation / double block and bleed where the owner procedure requires; gas-detection coverage verified before work; bonding/earthing. ADMIN: reinstatement procedure with independent verification; leak test before introduction of hydrocarbons; emergency response on standby for first fill. PPE: FR clothing and personal gas monitor where the site standard requires.	Process Engineer / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Line-opening permit; isolation certificate; leak-test / pre-startup records Review trigger / reference: Review on any change of system boundary or composition data. Ref: OSHA PSM (guidance) [S-09].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no introduction of hydrocarbons into new/modified systems without completed leak test, pre-startup sign-off and operations acceptance. Residual risk remains HIGH — acceptance only by the authorised owner representative under the owner risk matrix.
RA-03	Work where hydrogen sulfide (H2S) / toxic gas may be present — VERIFICATION-CRITICAL	Any process system — current gas composition NOT publicly confirmed [CONFIRM from process data]	Toxic-gas exposure if the actual stream is sour. Historical public reporting states the Qadirpur membrane plant technology was for CO2 and H2S removal [S-03] — that does NOT confirm current H2S content or concentration at any workfront.	Acute toxic exposure, single or multiple fatalities, failed rescue by unprotected responders. Persons at risk: Crews opening or working near process systems; exposure by inhalation at the work point and in low-lying areas.	NOT CONFIRMED: no current gas composition, P&ID or SDS data was available to this pre-assessment. Treat as unassessed until process safety information is supplied.	3 x 5 = 15 — HIGH	VERIFY FIRST: obtain current gas composition / H2S data, hazardous-area drawings and owner gas-testing criteria before any intrusive work. ENGINEERING (if sour service is confirmed): fixed H2S detection coverage, ventilation. ADMIN: H2S training for exposed crews, personal monitors with alarm limits set ONLY to the approved site standard (no values assumed here), buddy system, rescue plan with breathing-apparatus-equipped responders. PPE: escape respirators / BA as the	Process Safety / HSE Manager Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Owner gas-testing procedure; H2S contingency plan [SITE INPUT REQUIRED] Review trigger / reference: Review immediately when composition data is issued. Ref: historical process context [S-03]; OSHA PSM (guidance) [S-	Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: intrusive work on any process system is prohibited until composition data is received and the H2S-controls decision (required / not required) is recorded by a competent person.

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
							site standard requires.			09].	
RA-04	Isolation of mechanical / electrical / instrument energy for intrusive work (LOTO)	All systems in package [CONFIRM isolation points from P&IDs]	Failure to identify and isolate all energy sources (pressure, electrical, stored, gravity, chemical) — unexpected start-up or release during work.	Fatality or serious injury from pressure release, moving equipment, electric shock or falling load. Persons at risk: Fitters, electricians and instrument technicians at the work point; exposure via direct energy release.	TO BE VERIFIED: OGDCL isolation/LOTO procedure and isolation hardware availability not evidenced.	3 x 5 = 15 — HIGH	ENGINEERING: positive isolation (physical disconnection / blinding) for line breaking per owner procedure; lockable isolation points. ADMIN: isolation certificate listing every energy source, applied by an authorised isolator; personal locks/tags for each worker; zero-energy verification (try-start, test-for-dead, bleed check) recorded before work; group lockout for multi-craft work.	Authorised Isolator / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Isolation certificate; LOTO register; PTW Review trigger / reference: Review on any isolation boundary change. Ref: owner isolation/LOTO procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: work must not start until the zero-energy state is proven and recorded; any doubt about an energy source stops the job.
RA-05	Line opening / flange breaking, blinding/spading, joint making and reinstatement on process piping	Tie-in and modification points [CONFIRM from line list]	Residual pressure, trapped gas/liquid or incorrect line identification releases hydrocarbons or chemicals at the break point; line-of-fire ejection. Incorrect blind location, a passing isolation valve, an unrecorded blind left in / removed early, or incorrect gasket/bolt torque defeats the isolation or fails in service.	Fire, toxic exposure, fatality, environmental release; or delayed in-service joint failure, fire and unplanned shutdown. Persons at risk: Pipefitters at the flange face and personnel in the line of fire; exposure via jet release and spray.	TO BE VERIFIED: owner line-opening procedure, drain/vent provisions and lineup drawings not reviewed.	4 x 5 = 20 — EXTREME	ELIMINATION: break lines only at designated break points in the approved procedure. ENGINEERING: drain, vent, purge and prove the line depressurised at the break point before loosening; first-break technique from a protected position. ADMIN: line identification independently verified against the line list/P&ID by two competent persons; gas test at the break point; line-opening permit signed by operations. PPE: face/chemical protection per the fluid SDS and site standard. BLIND/JOINT CONTROLS: blind/spade register with unique tags, locations marked on P&IDs, installation/removal only by authorised persons with independent verification; joint register with QA/QC hold-point inspection before insulation/closure; controlled bolting with calibrated equipment and recorded values; specified gasket verified at fit-up; reinstatement leak test before handover.	Mechanical Supervisor / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Line-opening permit; isolation certificate; gas test certificate; blind register; approved WPS/ITP; joint register Review trigger / reference: Review for every new line break. Ref: owner line-opening procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: first break prohibited until drain/vent is proven, gas test is satisfactory and the permit is signed. Stop immediately on any unexpected pressure/product at the break. No repressurisation until the blind register is reconciled and signed by operations; no joint to be closed out before QA/QC hold-point release.
RA-06	Confined-space entry into vessels, tanks, sumps and enclosed/low-lying areas	Equipment and low points in package [CONFIRM entry list and classification]	Toxic/flammable/ oxygen-deficient atmosphere (including gas accumulating in low areas), engulfment or heat stress inside the space; failed rescue multiplying casualties.	Single or multiple fatalities (entrant and would-be rescuers). Persons at risk: Entrants, attendant and rescue team; exposure via the atmosphere inside the space.	TO BE VERIFIED: OGDCL confined-space procedure, equipment classification and rescue capability not evidenced.	3 x 5 = 15 — HIGH	ELIMINATION: use remote inspection/cleaning where practicable to avoid entry. ENGINEERING: isolation and blinding of all connected lines, cleaning/purging proven by testing, continuous ventilation and continuous atmospheric monitoring. ADMIN: entry permit with named attendant, communications check, entry log, and a rehearsed rescue plan with equipment at the entry point before entry starts; entrant/attendant training verified. PPE: respiratory protection per test results and site procedure.	Confined-Space Entry Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Confined-space entry permit; gas test certificate; rescue plan Review trigger / reference: Review for every entry. Ref: OSHA Confined Spaces overview (guidance only) [S-10]; owner procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no entry without rescue capability physically in place; entry stops immediately on monitor alarm, ventilation failure or the attendant leaving post. No entry where test results are outside the limits in the approved site procedure — limits are taken from that procedure, not

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
											assumed.
RA-07	Hot work: welding, cutting and grinding in or near process areas	Fabrication areas and in-situ hot-work locations [CONFIRM]	Sparks/slag igniting a flammable atmosphere, combustible materials or hydrocarbon residues; fire spreading beyond the work point.	Fire/explosion, fatalities, major plant damage. Persons at risk: Hot-work crew, fire watch, nearby workers and plant; exposure via ignition at and beyond the work point.	TO BE VERIFIED: OGDCL hot-work procedure, gas-testing criteria and fire-watch practice not reviewed.	3 x 5 = 15 — HIGH	ELIMINATION: prefabricate in a designated safe fabrication area wherever possible instead of in-situ hot work. ENGINEERING: spark containment (blankets/screens); removal/isolation of combustibles within the radius set by the site procedure. ADMIN: hot-work permit with pre-work and continuous gas testing where the procedure requires; dedicated trained fire watch during work and for the post-work fire-check period specified in the site procedure; suitable extinguishers at the point of work verified before starting. PPE: welding PPE per site standard.	Hot Work Supervisor / Fire Watch Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Hot-work permit; gas test certificate; fire-watch record Review trigger / reference: Review for every hot-work permit. Ref: OSHA welding/cutting guidance (guidance only) [S-11]; owner hot-work procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: hot work stops immediately on gas-test alarm, loss of fire watch, or any plant upset alarm in the area.
RA-08	Welding and thermal cutting — fume, radiation and burn exposure	All welding locations	Welding fume/metal-fume inhalation, UV/IR eye and skin damage, contact burns from hot metal; risk increases in enclosed spaces.	Occupational lung disease over time, eye injury, burns. Persons at risk: Welders and nearby helpers/bystanders; exposure via the fume plume and arc radiation.	TO BE VERIFIED: ventilation provisions, exposure monitoring and welder health surveillance not evidenced.	4 x 3 = 12 — HIGH	ENGINEERING: local exhaust ventilation / fume extraction for enclosed or semi-enclosed welding; welding screens to protect bystanders. ADMIN: fume exposure assessment for the consumables in use (from SDS) and monitoring where the assessment requires; hot pieces marked/quarantined until cool; rotation to limit individual exposure where monitoring indicates. PPE: welding helmet with correct shade, FR clothing, respiratory protection where ventilation cannot control fume (selected from the exposure assessment).	HSE Officer / Welding Supervisor Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	SDS file for consumables; exposure assessment Review trigger / reference: Review on consumable or location change. Ref: OSHA welding hazards (guidance only) [S-11].	Evidence status: To be verified — no site evidence reviewed STOP-WORK: welding in enclosed spaces stops if ventilation fails or monitoring exceeds the action level in the site hygiene plan.
RA-09	Electrical work, instrument work and temporary power supplies	Substations, field installations and construction areas [CONFIRM voltages from drawings]	Contact with live parts, backfeed from temporary generators/UPS/interconnections, arc flash, or damaged/trailing temporary cables in dusty or wet conditions.	Electrocution, arc-flash burns, fatality; fire from faulty temporary distribution. Persons at risk: Electricians, instrument technicians and all personnel using portable tools; exposure via direct contact, arc flash and damaged equipment.	TO BE VERIFIED: electrical safety rules, authorised-person register, drawings and temporary-power inspection regime not reviewed. Voltages are NOT stated in this assessment — obtain from approved drawings.	3 x 5 = 15 — HIGH	ENGINEERING: isolation at all sources including temporary supplies, earthing where the rules require; industrial-grade temporary distribution with RCD/earth-leakage protection and protected cable runs. ADMIN: work only by electrically authorised persons; prove dead with a verified tester before touching; electrical permit for work as defined by the site rules; temporary-supply register; inspection/tagging of portable tools with defective items removed. PPE: arc-rated clothing and voltage-rated gloves per the site electrical standard.	Authorised Electrical Person / E&I Engineer Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Electrical permit; isolation certificate; temporary installation certificates; inspection register Review trigger / reference: Review on any supply configuration change. Ref: owner electrical safety rules [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no contact work until prove-dead is recorded; any unidentified cable is treated as live and stops the job; any damaged cable takes the circuit out of service.
RA-10	Pressure testing — hydrostatic (and pneumatic only where specified and justified)	Test boundaries in package [CONFIRM test pressures from the approved test pack]	Failure of a test blank, hose, fitting or weak component under pressure: high-energy jet/projectile (hydrostatic) or violent stored-energy blast (pneumatic).	Fatality or serious injury; multiple fatalities possible in a pneumatic failure; equipment damage. Persons at risk: Test crew and anyone inside/beyond the test boundary; exposure	TO BE VERIFIED: test procedures, calibrated gauge records and boundary practice not reviewed. Test pressures are NOT stated here — they must come from the approved test pack.	3 x 5 = 15 — HIGH	ELIMINATION/ SUBSTITUTION: use hydrostatic testing wherever the specification allows; pneumatic testing only with a specific engineered risk review and senior written approval. ENGINEERING: test pack engineered and approved	Mechanical / Commissioning Engineer (Test Supervisor) Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved test pack; pressure-test permit; calibration certificates; pneumatic justification (if applicable) Review trigger /	Evidence status: To be verified — no site evidence reviewed HOLD POINT: pressurisation only under the Test Supervisor's single command with the

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
				via the line of fire and projectile/blast range.	Pneumatic testing is NOT CONFIRMED as in scope.		(medium, pressure, hold points, vents/drains) by a competent engineer; restraints on temporary hoses/connections; calibrated gauges with current certificates. ADMIN: barricaded exclusion zone around the full test boundary; staged pressurisation with inspection at each stage and remote monitoring for pneumatic steps; no one inside the boundary above the pressure level set in the test procedure; controlled depressurisation.			reference: Review for every test. Ref: approved test procedure [SITE INPUT REQUIRED].	boundary confirmed clear and recorded.
RA-11	Commissioning, energisation, first gas-in, nitrogen/inert-gas purging (if used) and handover to operations (incl. turnaround interfaces where in the work pack)	Systems under commissioning / turnaround [CONFIRM]	Leaks at unproven joints, incorrect lineup, unexpected energisation/rotation while other crews still work on connected systems, or open punch items transferred at handover. VERIFICATION-CRITICAL: if nitrogen/inert gas is used for purging, leakage or discharge can create an oxygen-deficient atmosphere without warning properties — use of N2 at the workfront is NOT confirmed in public sources.	Flammable/toxic release, fire, fatality, damage to operating plant; process incident at start-up. Persons at risk: Commissioning, operations and construction crews across the boundary; exposure at leak points and on shared systems.	TO BE VERIFIED: commissioning, pre-startup safety review (PSSR) and handover procedures are owner documents — not reviewed. No current turnaround is claimed in this assessment.	3 x 5 = 15 — HIGH	ENGINEERING: staged pressurisation with hold points and leak surveys at each stage. ADMIN: PSSR completed and signed before hydrocarbon introduction; formal system-boundary certificates with a single status register; energisation notices to all affected parties; permit systems cross-checked daily at the boundary meeting; punch items categorised — no start-up with open safety-critical items; exclusion zone during first fill sized by the commissioning procedure. NITROGEN (if used): purge plan with discharge routed to a safe location, barricaded vent points, marked exclusion zones, oxygen monitoring at access points and warning signage in workforce languages. PPE: per commissioning procedure.	Commissioning Lead / Operations Representative Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	PSSR certificate; commissioning permit; boundary and operations acceptance certificates; punch list Review trigger / reference: Review at each commissioning stage gate. Ref: owner PSSR procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no energisation of any system with open permits on that system; no hydrocarbon introduction without PSSR closure, leak-test records and signed operations acceptance; no entry into, or breaking into, any purged system until oxygen content is proven acceptable by test and recorded.
RA-12	Heavy and critical crane lifts; crane set-up on site ground; lifting in adverse weather	Laydown areas and plant installation points [CONFIRM]	Rigging failure, incorrect load weight/centre of gravity, or crane malfunction causing a dropped or swinging load. Ground-bearing failure / outrigger sinking into soft or uncompacted ground or over buried services causing crane overturn; loss of load control or lightning strike beyond approved weather limits.	Fatality, serious injury, damage to live plant with secondary hydrocarbon release. Persons at risk: Rigging crew and personnel in the drop zone; exposure via being under or near the suspended load path.	TO BE VERIFIED: OGDCL lifting procedure, crane/rigging certification and appointed-person arrangements not reviewed. OGDCL tender documents do place shifting/lifting responsibility on contractors for specific jobs [S-01] — that is not evidence of current lifting arrangements.	3 x 5 = 15 — HIGH	ENGINEERING: lift only with certified crane and rigging matched to a calculated lift plan (load weight and CoG from vendor/drawing data — never estimated); tag lines for load control. ADMIN: lifts categorised per the owner lifting standard with an approved lift plan and appointed person for complex/critical lifts; barricaded drop zone; trial lift and brake-hold check; no personnel under suspended loads, enforced by the lift supervisor. GROUND/WEATHER: engineered crane pads/mats sized from a ground-bearing assessment by a competent person, outrigger loads calculated in the lift plan; underground-service clearance before positioning; daily ground check and re-verification after rain; set-up sign-off by the appointed person before the first lift; wind/weather limits from	Lifting Supervisor / Appointed Person Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved lift plan (with ground assessment); crane/rigging certificates; utility clearance for pad area; weather-monitoring log; lifting permit where required Review trigger / reference: Review for every complex/critical lift. Ref: owner lifting procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: critical lifts over or near live plant require operations approval in addition to the lift plan; no lift from any set-up position without signed ground/set-up verification — any visible settlement stops lifting; a lift stops on any rigging defect, alarm or loss of communication. Any worker may stop work on lightning or a sudden dust storm.

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
							the crane manual and the approved site procedure (no numeric limits assumed here), anemometer readings at the workfront, and defined stop-work triggers for lightning and dust-storm visibility. PPE: task PPE.				
RA-13	Scaffold erection/alteration/dismantling, work at height and MEWP use; dropped objects	Elevated work areas in package [CONFIRM]	Scaffold collapse from incorrect erection/ties/loading; fall from height through open edges, penetrations or fragile surfaces; tools/materials dropped onto personnel below. MEWP overturn on soft/uneven ground, ejection/entrapment of occupants, or contact with overhead structures or live equipment.	Fatality or life-changing injury to the faller, scaffolder or the person struck. Persons at risk: Scaffolders, fitters, welders and inspectors at height; all personnel at lower levels.	TO BE VERIFIED: scaffold standard, inspection/tag system, anchor certification, scaffolder competency and rescue arrangements not evidenced.	4 x 5 = 20 — EXTREME	ELIMINATION: pre-assemble at grade where practicable. ENGINEERING: scaffolds erected to an approved design/standard with ties, bracing and loading limits, and a design check for special scaffolds; collective protection first — guardrails, covered/marked penetrations, certified anchor points; tool tethering, toe boards and barricaded drop zones below. ADMIN: erection/alteration only by competent scaffolders under a scaffold supervisor; inspection and tag before first use, periodically and after alteration or severe weather, with incomplete scaffolds physically barred; work-at-height permit where the site system requires; rescue-from-height plan and equipment at the workfront before work starts. MEWPs: machines selected for terrain and reach, inspected and certified, ground assessed before positioning (RA-12 principles); trained/authorised operators only; recorded pre-use checks; exclusion zone and ground spotter near plant; emergency lowering procedure briefed. PPE: full-body harness with double lanyard per site standard where required.	Scaffold Supervisor / Construction Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Scaffold inspection register and tag; design drawings for special scaffolds; WAH permit (if required); rescue plan; MEWP certificates and operator authorisations Review trigger / reference: Review periodically and after any storm. Ref: owner scaffold/WAH procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no access to any scaffold without a current 'safe for use' tag; work at unprotected edges is prohibited; any fall-arrest deployment or anchor defect stops the task pending review. Defective/uncertified MEWPs are tagged out; no MEWP operation outside the manufacturer's wind limit (from the machine manual — not assumed here).
RA-14	Excavation and trenching for foundations, pipelines and underground services	Civil work areas in package [CONFIRM]	Trench/excavation wall collapse; strike on buried live services not shown or incorrectly marked; falls into excavations; spoil/plant surcharge at the edge.	Burial/asphyxiation fatality; electrocution or hydrocarbon release from a service strike; fractures from falls. Persons at risk: Workers in and beside the excavation, plant operators and all personnel near unprotected edges.	TO BE VERIFIED: underground-service drawings, permit practice, soil data and edge-protection provisions not reviewed. OGDCL tender scope for Qadirpur does include foundation excavation work of this type [S-01] — quantities in that tender are NOT used here.	3 x 5 = 15 — HIGH	ENGINEERING: protective system (benching/shoring/shielding) selected by a competent person for the actual soil and depth — dimensions from the excavation design, not assumed here; service-detection scan and trial holes to prove buried-service positions before mechanical digging; rigid edge barriers, safe crossing points and access/egress. ADMIN: excavation permit with services clearance signed by operations/E&I; daily and after-rain inspections by the competent person; mechanical excavation stops at the agreed distance from marked services and hand digging takes over; spoil and plant kept back at the	Civil Supervisor / Competent Person (Excavations) Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Excavation permit; services-clearance drawings; scan records; inspection records Review trigger / reference: Review daily and after rain or design change. Ref: owner excavation procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no mechanical excavation without signed services clearance; any unmarked service found, or any failed edge protection, stops/closes the excavation until corrected.

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
							distance set by the design; open excavations barricaded, signed and lit overnight.				
RA-15	Site traffic, mobile plant and driving on access/public roads near Ghotki	Site roads, laydown areas and access routes [CONFIRM traffic plan]	Vehicle-pedestrian collision, reversing in blind spots, collision/rollover on access roads aggravated by fatigue, dust, speed and mixed public traffic.	Fatality or serious injury to pedestrians, drivers, passengers and other road users. Persons at risk: Pedestrians in plant areas, drivers, passengers and the public on shared access routes.	TO BE VERIFIED: traffic management plan, driver authorisation and journey management not reviewed.	4 x 5 = 20 — EXTREME	ENGINEERING: segregated pedestrian walkways and crossings; one-way systems and physical separation in laydown areas; reversing alarms/cameras where fitted. ADMIN: site traffic plan with speed limits set by the plan (values from the plan — not assumed); authorised-driver register; banksman for reversing in congested areas; journey management with rest requirements for longer drives; seat-belt rule for all occupants; no night driving on remote routes unless approved. PPE: high-visibility clothing in plant/traffic areas.	Construction Manager / Logistics Coordinator Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Traffic management plan; driver authorisation register; journey plans Review trigger / reference: Review monthly and after any traffic incident/near miss. Ref: site traffic plan [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed STOP-WORK: any vehicle defect affecting brakes/steering/alarms removes the vehicle from service; pedestrian entry into an active plant-movement zone stops the movement.
RA-16	Outdoor work in extreme heat (Sindh summer) — heat stress, dehydration, heatstroke	All outdoor workfronts — Ghotki, Sindh	Heat exhaustion progressing to heatstroke during high-temperature work; risk peaks in the Sindh summer, during heavy physical work and for unacclimatised workers.	Heatstroke can be fatal or cause permanent organ damage; multiple casualties possible on peak days. Persons at risk: All outdoor workers, especially new arrivals, labourers, scaffolders and PPE-intensive trades; exposure via ambient heat plus metabolic load.	VERIFIED LEGAL FRAMEWORK: the Sindh Occupational Safety and Health Act, 2017 (Sindh Act No. I of 2018) and the Sindh OSH Rules, 2019 place general employer duties for worker safety and health in Sindh [S-07] — exact current texts to be confirmed in the project legal register. The site heat-stress programme itself is NOT evidenced — to be verified. NOTE: no Saudi-style midday-work ban applies in Pakistan; do not import one.	4 x 4 = 16 — HIGH	ENGINEERING: shaded rest areas at each workfront; cooled drinking water and electrolytes freely available at the point of work. ADMIN: a site heat-stress programme under the Sindh OSH framework and the OGDCL HSE system — acclimatisation schedule for new/returning workers; work/rest cycles driven by measured conditions (e.g. WBGT) under the site procedure; rescheduling of the heaviest work to cooler hours where the programme requires; buddy checks and supervisor training to recognise heat illness; rehearsed emergency response for suspected heatstroke (rapid cooling and medical evacuation). PPE: breathable/light PPE selections where the task allows.	HSE Manager / Contractor Supervisors Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 4 = 8 — MODERATE	Site heat-stress procedure; heat-illness response plan Review trigger / reference: Review at season start and after any heat-illness case. Ref: Sindh OSH Act 2017 / Rules 2019 [S-07]; site procedure [SITE INPUT REQUIRED].	Evidence status: Legal framework verified; site programme to be verified STOP-WORK: any suspected heatstroke case stops the crew's work, triggers medical response and a programme review; work stops at the measured-condition limits set in the site heat-stress procedure.
RA-17	Cutting, drilling and concrete work — respirable dust/silica; noise and hand-arm vibration	Civil, mechanical and fabrication areas [CONFIRM exposure survey]	Respirable crystalline silica from concrete/stone cutting and drilling; excessive noise from plant and tools; hand-arm vibration from breakers/grinders.	Silicosis and lung disease (chronic), noise-induced hearing loss, hand-arm vibration syndrome. Persons at risk: Operators and nearby workers; exposure via dust plumes, noise zones and tool vibration over repeated shifts.	TO BE VERIFIED: no exposure monitoring data exists in public sources — an industrial-hygiene survey is required before ratings can be confirmed.	4 x 3 = 12 — HIGH	ENGINEERING: wet cutting / on-tool extraction for silica tasks; low-noise/low-vibration tools selected at procurement; noise barriers/distance for fixed plant. ADMIN: exposure assessment and monitoring programme by a competent hygienist; hearing-protection zones signed where monitoring requires; tool-use time limits and rotation for vibration; health surveillance for exposed groups. PPE: fit-tested respiratory protection for silica tasks and hearing protection in signed zones (selected from monitoring results).	HSE Manager / Occupational Hygienist [competent person] Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	Industrial hygiene survey and monitoring plan Review trigger / reference: Review on monitoring results and task changes. Ref: site hygiene plan [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed STOP-WORK: dry cutting of concrete/stone is prohibited where wet/extraction controls are unavailable.
RA-18	Manual handling, housekeeping, slips, trips and falls	All work areas, walkways,	Musculoskeletal injury from handling valves, blinds and	Strains, fractures and lost-time injuries; serious incidents	TO BE VERIFIED: housekeeping inspection regime,	4 x 3 = 12 — HIGH	ENGINEERING: mechanical handling aids planned into tasks; cable/hose ramps and	Contractor Supervisors / Construction	2 x 3 = 6 — MODER	Manual-handling assessments; housekeeping	Evidence status: To be verified — no site evidence

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
	on the level; night work and fatigue	stores and shifts	scaffold components; trips over cables/hoses/debris; slips on spilled liquids or sand; fatigue-related errors and misunderstood instructions on night/extended shifts.	during low-supervision periods. Persons at risk: All trades on all shifts; exposure via routine movement/handling and reduced alertness at night.	mechanical-aid availability, shift/fatigue rules, lighting levels and briefing-language arrangements not evidenced.		designated walkways kept clear; task lighting to the site standard at night workfronts. ADMIN: task-specific manual-handling assessment for heavy/awkward items with team-lift rules; housekeeping standard with daily area sign-off; immediate-clean rule for spills; fatigue-management rules in the site plan (maximum hours and rest — values from the plan/owner standard, not assumed); toolbox talks and permits briefed in the languages the crew actually understands, with comprehension checks; high-risk work restricted to day shift unless a specific night-work assessment is approved. PPE: safety footwear and gloves per task.	Manager Priority: Moderate Target date: [SITE INPUT REQUIRED]	ATE	inspection records; fatigue management plan; lighting survey Review trigger / reference: Review weekly via inspection trends and monthly for fatigue. Ref: site HSE plan [SITE INPUT REQUIRED].	reviewed STOP-WORK: walkways/escape routes found blocked are cleared before work continues in that area. HOLD POINT: critical lifts, line opening, confined-space entry and first energisation are day-shift-only unless a night-work risk assessment is approved by the Construction and HSE Managers.
RA-19	Use of chemical products: coatings, solvents and cleaning/insulation materials — SDS-DEPENDENT	Coating/ maintenance locations [CONFIRM product list]	Flammable/toxic product exposure, skin/respiratory damage, fire from solvent vapours or environmental release — specific hazards depend entirely on the actual products.	Occupational illness, fire, spill contamination. Persons at risk: Applicators, maintenance crews and nearby workers; exposure via inhalation, skin contact and ignition sources.	NOT CONFIRMED: no product list or SDS file was available to this pre-assessment.	3 x 3 = 9 — MODERATE	VERIFY FIRST: obtain the full product list and current SDS for every product before mobilisation. ENGINEERING: ventilated application/mixing areas; flammables stored in approved cabinets/containers away from ignition. ADMIN: product-specific risk assessment per SDS, crew briefing in their language, spill kits at use points, waste disposed via the site waste system. PPE: strictly per SDS (gloves, eye protection, respirator type as specified).	HSE Officer / Contractor Supervisor Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	SDS file; product-specific assessments; hot-work controls near flammables Review trigger / reference: Review on every new product. Ref: product SDS [SITE INPUT REQUIRED before use].	Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: no product may be used on site before its SDS assessment is complete and briefed to the crew.
RA-20	NDT / industrial radiography — CONDITIONAL (only if radiography is in the work pack)	Weld inspection locations [CONFIRM]	Ionising radiation exposure from an unshielded or uncontrolled source; source loss or failure to retract.	Radiation overexposure of the radiography crew or unknowing bystanders; regulatory incident. Persons at risk: Radiography crew and any person entering the controlled area; exposure via proximity to the exposed source.	NOT CONFIRMED: use of radiography at the workfront is not evidenced in public sources — confirm from the inspection plan/ITP and the applicable Pakistan regulatory licensing route (not assumed here).	2 x 4 = 8 — MODERATE	ENGINEERING: collimation and shielding to minimise the controlled area. ADMIN: radiography only by licensed/authorised personnel under the applicable Pakistan requirements and site procedure; calculated and barricaded controlled area with warning signs/lights in workforce languages; personal dosimetry and survey-meter checks before, during and after exposure; source accountability log reconciled every shift. PPE: dosimetry (monitoring, not protection).	NDT Supervisor / Radiation Protection Officer [role to be confirmed] Priority: High Target date: [SITE INPUT REQUIRED]	1 x 4 = 4 — LOW	Radiography permit; controlled-area calculation; dosimetry records Review trigger / reference: Review for every radiography operation. Ref: approved ITP and radiation procedure [SITE INPUT REQUIRED].	Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: no exposure while any person is inside the controlled area or the survey meter is unavailable/defective.
RA-21	Emergency response interface (plant fire / gas emergency) and severe weather: monsoon rain/flooding, lightning and dust storms	Whole defined plant boundary	Maintenance/ construction personnel not aligned with the plant emergency plan — wrong muster point, missed headcount, re-entry into a hazardous area, or blocked emergency access during a plant alarm.	Multiple fatalities in a major fire/gas event; delayed rescue. In severe weather: drowning/entrapment in flooded excavations, lightning fatalities, collapse of temporary works and vehicle incidents. Persons at risk: Every person inside the	TO BE VERIFIED: the approved site emergency plan, alarm signals, muster points, medical provision and headcount system were NOT available to this pre-assessment — all details are critical site inputs, none are assumed here. Unverified third-	2 x 5 = 10 — HIGH	ADMIN: every worker inducted on the actual alarm signals, muster points and wind-direction rule from the approved plan before starting work; contractor headcount reconciled with the plant muster system (gate/badge data — see Section 5); joint emergency drills including maintenance/construction crews at the frequency set by	HSE Manager / Plant Emergency Coordinator [role to be confirmed] Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved site emergency response plan; drill records; induction records Review trigger / reference: Review after every drill, alarm event, workforce change or severe-weather	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no workfront may operate outside alarm coverage or without a briefed muster route; a failed drill triggers corrective action

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
				boundary, especially new and transient workers; exposure via the plant event itself and failed evacuation.	party student-report claims about muster arrangements at Qadirpur were NOT used.		the plan; emergency access routes kept clear and audited; wind socks maintained and visible at workfronts. ENGINEERING: alarm coverage audible/visible at all workfronts including high-noise areas (beacons). SEVERE WEATHER (Sindh monsoon/lightning/dust): weather monitoring with defined alert triggers and a named decision-maker per shift (triggers from the site procedure — not assumed); pre-storm securing checklist for cranes, scaffolds, temporary structures and loose materials; excavations and low areas evacuated and closed on a heavy-rain warning and inspected before re-entry; no driving through flooded sections; lightning shelter provision per the site plan.			event, and at season changes. Ref: site emergency plan and severe-weather procedure [SITE INPUT REQUIRED — critical].	before the next high-risk activity. STOP-WORK: all outdoor work stops on a lightning or flood warning; re-entry to excavations only after competent-person inspection (links to RA-14).
RA-22	Hydrocarbon/ chemical handling, refuelling, condensate and produced-water interfaces, waste and spill response	Refuelling points, laydown, work areas and drainage paths [CONFIRM]	Spills of fuel, condensate, chemicals or contaminated water reaching soil or drains; incorrect waste segregation (including contaminated PPE/absorbents) creating fire and environmental harm. Produced-water handling interfaces apply only where the work pack confirms them.	Soil/groundwater contamination, fire, regulatory liability, harm to the local environment. Persons at risk: Refuelling and maintenance crews; the environment; exposure via direct contact and contaminated ground.	TO BE VERIFIED: site environmental management plan, waste arrangements and drain/seal status not reviewed. Applicable Pakistan/Sindh environmental requirements to be confirmed in the project legal register — none are cited from memory.	3 x 3 = 9 — MODERATE	ENGINEERING: bunded fuel/chemical storage sized to the site standard; refuelling on impermeable surfaces with drip trays; sealed/blanked drains in chemical areas where the design requires. ADMIN: spill kits at storage and refuelling points with trained responders; spill reporting and clean-up procedure in the environmental plan; waste segregation at source with labelled containers and a licensed/authorised disposal route per the current applicable requirements (to be confirmed). PPE: per SDS.	Environmental Officer / HSE Officer Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	Site environmental management plan; waste transfer records Review trigger / reference: Review monthly and after any spill. Ref: site environmental plan [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed STOP-WORK: any spill reaching open soil or a drain stops the activity until contained and reported under the environmental plan.

NOTE ON RESIDUAL RATINGS: several major-accident rows (e.g. RA-02, RA-03, RA-05) retain HIGH residual scores because severity cannot be reduced — only likelihood can. Those risks are acceptable, if at all, only through the owner's formal acceptance route with all stated controls verified in place. PPE alone never justifies a rating reduction in this register.

8. Critical controls / verification checklist

Each critical barrier for a high-consequence event must be verified with evidence before the associated work proceeds. 'To be verified' means exactly that — the barrier is not credited until the evidence exists and a competent verifier has signed it.

#	Critical control / barrier	Applicability	Evidence required	Verifier (role)	Frequency	Verification status
1	Live-plant / SIMOPS interface review completed and work authorised by operations for the area and shift	To be verified	Signed SIMOPS review / permit coordination record	Area Authority	Each shift / each new workfront	Not verified
2	Process isolation / positive isolation proven and LOTO applied before intrusive work	To be verified	Isolation certificate with zero-energy verification recorded	Authorised Isolator	Every intrusive task	Not verified
3	Line-opening controls: drain/vent/purge proven, first-break technique, gas test at break point	To be verified	Line-opening permit + gas test certificate	Mechanical Supervisor	Every line break	Not verified
4	Gas composition / H2S decision recorded before intrusive work (RA-03)	To be verified — composition data required	Composition data sheet; signed H2S-controls decision note	Process Safety / HSE Manager	Before any intrusive work	Not verified
5	Gas testing before and during work where the owner procedure requires it	To be verified	Gas test certificates; tester authorisation and monitor calibration records	Authorised Gas Tester	Per permit / continuous where	Not verified

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#	Critical control / barrier	Applicability	Evidence required	Verifier (role)	Frequency	Verification status
					required	
6	Hot-work permit, ignition control, trained fire watch and post-work fire check in place	To be verified	Hot-work permit; fire-watch log; extinguisher checks	Hot Work Supervisor	Every hot-work task	Not verified
7	Confined-space controls: classification, isolation, testing, attendant, communications, rescue plan with equipment at the entry point	To be verified	Entry permit; test records; rescue drill/equipment records	Entry Supervisor	Every entry	Not verified
8	Lifting controls: approved lift plan, certified crane/rigging, ground-bearing verification, barricaded drop zone, appointed person for complex/critical lifts	To be verified	Lift plan; certificates; set-up sign-off	Appointed Person (Lifting)	Every complex/critical lift; daily for routine	Not verified
9	Work-at-height controls: collective protection, inspected/tagged scaffolds, fall arrest with verified anchors, rescue plan at the workfront	To be verified	Scaffold tags/register; harness inspection records; rescue plan	Scaffold Supervisor / HSE Officer	Daily checks; periodic scaffold inspection	Not verified
10	Excavation controls: permit, services clearance and scan, protective system, access/egress, daily competent-person inspection	To be verified	Excavation permit; scan records; inspection log	Competent Person (Excavations)	Daily and after rain	Not verified
11	Electrical controls: authorised persons only, prove-dead recorded, temporary-power inspection and tagging	To be verified	Electrical permits; prove-dead records; inspection register	Authorised Electrical Person	Every task; periodic installation inspection	Not verified
12	Pressure-test controls: approved test pack, calibrated gauges, barricaded exclusion zone, single command, controlled depressurisation	To be verified	Test pack; calibration certificates; boundary-clear record	Test Supervisor	Every test	Not verified
13	Emergency readiness: alarm coverage at all workfronts, briefed muster points, headcount reconciliation, joint drills, clear emergency access	To be verified — emergency plan is a critical site input	Induction and drill records; muster reconciliation reports	HSE Manager	Drills per plan; headcount every shift	Not verified
14	Heat-stress controls: water/shade/rest at workfronts, acclimatisation, measured-condition work/rest, heatstroke response (Sindh summer)	Applicable — Sindh OSH legal framework verified [S-07]; site programme to be verified	Heat-stress procedure; condition/WBGT logs; training records	HSE Manager	Daily in hot season	Not verified
15	Contractor competency: verified qualifications/authorisations, induction and toolbox talks in workforce languages, supervision in place	To be verified	Competence matrix; induction records; authorisation registers	Construction Manager / HSE Manager	At mobilisation; periodic audit	Not verified
16	Environmental containment: bunded storage, spill kits and trained responders, drain protection, waste segregation	To be verified	Inspection records; waste transfer notes; spill log	Environmental Officer	Periodic inspection	Not verified
17	Management of change / impairment control: no bypass or override of safety-critical systems without approved MOC/impairment permit and compensating measures	To be verified	Bypass/impairment register; MOC records	Process Safety Engineer	Reviewed every shift while active	Not verified

9. Task sequencing / JSA / permit matrix

Permit names below are generic functional descriptions. The OGDCL permit-to-work system governs the actual permit types, forms and approval chain — obtain the controlled site PTW procedure (Section 14) and re-map this table to it before use. A task-level JSA/JHA must be produced for each activity step by the executing contractor and briefed to the crew before work.

Task / activity	Prerequisite permit(s)	Isolation requirements	Drawings / documents	Competent persons	Pre-job briefing	Field verification	Return to service / handover
Work adjacent to live process equipment	OGDCL PTW; SIMOPS authorisation	As defined by operations for the task	Plot plan extract; method statement; this RA	Area Authority; Contractor Supervisor	Joint operations/contractor toolbox talk	Area Authority field check before start	Permit close-out with operations
Hot work in/near process areas	Hot-work permit; PTW	Equipment isolation as required by permit	Gas test certificate; fire-watch assignment	Hot Work Supervisor; Fire Watch; Gas Tester	Permit briefing at the work point	Gas test before and during work; fire-watch log	Post-work fire check signed; permit closed
Line opening / flange breaking	Line-opening permit; PTW	Positive isolation + drain/vent/purge proven	P&ID; line list; isolation certificate; blind register	Authorised Isolator; Mechanical Supervisor	First-break briefing incl. line-of-fire positions	Two-person line ID verification; gas test at break	Reinstatement checklist; blind register reconciled
Confined-space entry	Entry permit; PTW	Blinding/isolation of all connected lines; lockout of agitators etc.	Entry plan; rescue plan; gas test records	Entry Supervisor; Attendant; Gas Tester; Rescue Team	Entrant/attendant briefing and comms check	Continuous monitoring; entry log maintained	All persons/equipment accounted for; permit closed
Excavation near services	Excavation permit	Service isolation where required by owner	Services drawings; scan records; excavation design	Competent Person (Excavations)	Crew briefing on marked services	Daily inspection; hand-dig zone enforcement	Backfill/reinstatement record
Complex / critical lifting	Lifting permit (if required by site system)	Area cleared; plant isolation if over live equipment (operations decision)	Approved lift plan; crane/rigging certificates; ground assessment	Appointed Person; Crane Operator; Riggers; Signaller	Lift briefing incl. abort signals	Trial lift; drop-zone and weather check	Lift completion report; defects logged
Electrical work	Electrical permit (as	Isolation at all sources;	Single-line diagrams;	Authorised Electrical	Briefing on boundaries	Prove-dead recorded at	Restoration under

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Task / activity	Prerequisite permit(s)	Isolation requirements	Drawings / documents	Competent persons	Pre-job briefing	Field verification	Return to service / handover
	defined by site rules)	earthing per rules	isolation certificate	Person	and backfeed sources	the work point	permit; records updated
Pressure testing	Pressure-test permit	Test boundary isolation; blinds per test pack	Approved test pack; calibration certificates	Test Supervisor; Mechanical Engineer	Boundary briefing; exclusion rules	Boundary-clear confirmation before pressurisation	Depressurisation record; test certificate
Commissioning / energisation / first gas-in	Commissioning permit; energisation notice	Boundary isolation per boundary certificate	PSSR records; boundary certificates; punch list	Commissioning Lead; Operations Representative	Boundary meeting with all affected parties	Permit cross-check before energisation	Operations acceptance certificate
Nitrogen purging (if used)	Commissioning/ maintenance permit	Isolation of purge boundary	Purge plan; gas test records	Commissioning Lead; Gas Tester	Briefing on asphyxiation risk and exclusion zones	O2 monitoring at access points	Oxygen proven normal and recorded before handover

10. Responsible-role matrix

Accountability is assigned to ROLES, never to invented individuals. Role titles below are functional descriptions; the site organisation chart and the OGDCL PTW procedure must confirm the actual titles, appointments and authority before this document is approved.

Role	Accountability in this assessment	Authority / confirmation note
Area Authority / Operations Representative	Owns the plant area; authorises work and SIMOPS; first contact for plant upsets	Title and authority to be confirmed against the site organisation chart and OGDCL PTW procedure
Permit Issuer / Permit Receiver (as defined by the site PTW system)	Issue, acceptance, suspension and close-out of permits within their competence	Exact titles and authorisation route per the controlled site PTW procedure — not assumed
Plant / Field Manager [site title to be confirmed]	Overall plant operation and the operating/maintenance interface	Confirm the current site management structure — not publicly disclosed
Maintenance / Construction Manager	Delivery, resources, sequencing and interface management for maintenance/construction work	Approves high-risk work scheduling with HSE and operations
Contractor Supervisor (each contractor)	Direct supervision of their crews; toolbox talks; first-line control verification	One named supervisor per crew per shift — names from contractor mobilisation records
HSE Manager / HSE Officer	RA ownership, inspections, incident reporting, heat-stress and hygiene programmes, drill coordination	Stop-work authority to be confirmed in the site HSE plan
Process Safety Engineer	Major-accident hazard interface, MOC/impairment control, PSSR participation	Confirms where HAZID/HAZOP/LOPA studies are required
Mechanical Engineer / Supervisor	Piping, line opening, pressure testing and joint integrity	Approves test packs jointly with commissioning/operations
Electrical / Instrumentation Engineer	Electrical safety, isolations, loop testing and energisation readiness	Authorised-person register maintained by this function
Lifting Supervisor / Appointed Person	Lift planning, categorisation, set-up verification and lift execution control	Appointed in writing per the owner lifting standard
Authorised Gas Tester	Atmospheric testing for permits, entries and hot work	Authorisation and calibration records current
Confined-Space Attendant / Rescue Team	Entry control, communications and rescue execution	Rescue team trained and equipped at the entry point
QA/QC Inspector	Hold-point inspection for joints, welds, coating and reinstatement	Releases hold points in the ITP only
Commissioning Lead	System completion, PSSR, energisation and handover boundaries	Single accountable authority at the maintenance/operations boundary
Emergency Response Coordinator [site title to be confirmed]	Alarm response, muster/headcount reconciliation, drill programme with plant operations	Interface defined in the approved site emergency plan

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11. Action tracker

No action is closed merely because it is written here. Closure requires the stated verification evidence and sign-off by the closure reviewer. Overdue Critical actions block the associated work (see hold points in Section 7).

Action ID	Linked RA	Action (specific)	Priority	Accountable role	Target	Verification evidence	Status	Closure reviewer
ACT-01	RA-01/02	Obtain approved plot plan and fix the assessment boundary, workfront list and asset IDs; re-issue this RA against them	Critical	Operations / Document Control	Before approval	Plot plan revision on file; RA Rev. 1 boundary table	Open	HSE Manager
ACT-02	RA-02/03	Obtain current gas composition data (incl. H2S and other hazardous constituents) and record the H2S-controls decision for every system in the package	Critical	Process Safety Engineer	Before any intrusive work	Composition data sheet; signed decision note	Open	HSE Manager
ACT-03	All	Replace the proposed 5x5 matrix with the OGDCL risk matrix and re-rate all 22 entries in a validation workshop	Critical	HSE Manager (owner)	Before approval	Workshop record; re-rated register	Open	Plant / Field Manager
ACT-04	All	Obtain current authorized OGDCL HSE/PTW/LOTO/hot-work/confined-space/line-opening/excavation/lifting/ pressure-test/MOC/emergency procedures and re-map Sections 8-9 to them	Critical	Document Control (owner)	Before approval	Controlled copies in document register	Open	HSE Manager
ACT-05	RA-21	Obtain the approved site emergency plan; extract alarm signals, muster points, routes, medical and rescue provisions into the induction pack	Critical	HSE Manager	Before work starts	Emergency plan on file; induction content updated	Open	Operations Representative
ACT-06	RA-21 / Sec.5	Start daily actual-headcount capture (gate/badge + contractor returns) and reconcile with the muster system; populate Section 5.2	Critical	Contractor Admin / Security	Before work starts; then daily	Dated manpower reports; reconciliation log	Open	HSE Manager
ACT-07	RA-04/05	Verify isolation hardware, blind register practice and authorised-isolator appointments by field inspection	Critical	Area Authority	Before first line break	Inspection report; appointments in writing	Open	Mechanical Engineer
ACT-08	RA-12	Complete ground-bearing assessments and crane-pad verification for all planned critical-lift set-up positions	High	Appointed Person (Lifting)	Before first critical lift	Signed set-up verifications	Open	Maintenance / Construction Manager
ACT-09	RA-16	Confirm the heat-stress programme: site procedure under the Sindh OSH framework, measured-condition monitoring, water/shade/rest provision, acclimatisation roster	Critical	HSE Manager	Before hot season / immediately for current work	Programme document; condition logs started	Open	Maintenance / Construction Manager
ACT-10	RA-14	Complete underground-services survey/scan for all excavation areas in the first work package	Critical	Civil Supervisor / Operations	Before excavation	Scan records; marked drawings	Open	Maintenance / Construction Manager
ACT-11	RA-06	Prove confined-space rescue capability (trained team + equipment) by a practical drill at the first entry location	Critical	Entry Supervisor	Before first entry	Drill report with corrective actions	Open	HSE Manager
ACT-12	RA-10	Approve test packs (hydrostatic default; pneumatic only with engineered justification) and verify gauge calibration certificates	High	Mechanical / Commissioning Engineer	Before first test	Approved test packs; certificates	Open	Commissioning Lead
ACT-13	RA-13	Inspect and tag all existing scaffolds in the work area; close out red-tag items; verify anchor points for fall arrest	High	Scaffold Supervisor	Before work at height	Scaffold register updated	Open	HSE Officer
ACT-14	RA-15	Issue the site traffic plan (routes, walkways, speed limits, driver authorisations) and brief all drivers	High	Logistics Coordinator	Before peak work	Plan issued; briefing records	Open	Maintenance / Construction Manager
ACT-15	RA-17	Commission an industrial-hygiene baseline survey (silica, noise, vibration, welding fume) for the package tasks	Moderate	HSE Manager	Within first month of work	Survey report; monitoring plan	Open	HSE Manager
ACT-16	RA-19/08	Collect the full product/SDS file for coatings, solvents and insulation materials; complete product-specific assessments	High	HSE Officer	Before product mobilisation	SDS file; assessment records	Open	Contractor Supervisor
ACT-17	RA-11	Audit the live bypass/impairment register for the interface areas; close or formally extend every overdue item	Critical	Process Safety Engineer	Before hot/intrusive work	Register audit report	Open	Operations Representative
ACT-18	All	Conduct the site walkdown and worker consultation for this assessment; record observations and amend the register; confirm current plant configuration against P&IDs (historical public descriptions must not be relied on)	Critical	HSE Manager	Before approval	Walkdown report; consultation records; RA Rev. 1	Open	Plant / Field Manager

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12. Emergency response and rescue interface

The approved site emergency response plan was NOT available to this pre-assessment. Nothing in the table below is assumed: every operational detail is a critical site input to be extracted from that plan and briefed at induction (links to RA-21 and ACT-05). Unverified third-party claims about Qadirpur muster arrangements found during research were deliberately NOT used.

Emergency element	Detail required from the approved site emergency plan	Status
Emergency number / control-room contact	Exact number(s) and radio channels for raising the alarm from any workfront	SITE INPUT REQUIRED — critical
Alarm signals	Signal types and meanings for fire, gas release, evacuation and all-clear; audibility/visibility at high-noise workfronts	SITE INPUT REQUIRED — critical
Muster points	Locations per area, wind-direction alternates, and the route to each from every workfront	SITE INPUT REQUIRED — critical
Headcount / muster reconciliation	System used (badge/gate data), who reconciles contractor numbers, and the reporting line to the incident commander	SITE INPUT REQUIRED — critical
Wind indication	Wind-sock locations and the rule for selecting upwind muster routes	SITE INPUT REQUIRED — critical
Medical provision	Clinic location [CONFIRM], ambulance provision and route to the receiving hospital [CONFIRM] — no response times are stated until evidenced	SITE INPUT REQUIRED — critical
Fire response	Plant fire team interface, hydrant/firewater status at the interface, extinguisher provision for work areas	SITE INPUT REQUIRED — critical
Gas-release response	Actions on gas-detector alarm, safe refuge locations and re-entry authority	SITE INPUT REQUIRED — critical
Rescue — confined space	Team, equipment and drill standard (see RA-06)	SITE INPUT REQUIRED — critical
Rescue — work at height	Equipment and trained rescuers per workfront (see RA-13)	SITE INPUT REQUIRED — critical
Communications dead zones	Radio/mobile coverage survey for remote fronts and the fallback check-in system	SITE INPUT REQUIRED — critical
Contractor–operator interface	Who commands, who accounts for contractor personnel, and how permits are suspended on alarm	SITE INPUT REQUIRED — critical

13. Environmental and occupational-health screening

Screening only: no exposure level is stated anywhere in this document, because no monitoring data exists in public sources. Where the screening identifies a credible exposure, an industrial-hygiene assessment/monitoring action is raised (ACT-15).

Aspect	Source activity (register link)	Screening conclusion	Control / monitoring requirement
Dust / respirable silica	Concrete cutting, drilling, excavation (RA-14, RA-17)	Credible exposure — unquantified	Wet cutting/extraction; hygiene survey and monitoring; fit-tested RPE where required
Welding fume / metal fume	Welding and thermal cutting (RA-08)	Credible exposure — unquantified	LEV/extraction; consumable SDS assessment; monitoring in enclosed spaces
Noise	Plant, compressors, tools, generators (RA-17)	Credible exposure — unquantified	Noise survey; signed hearing-protection zones from results
Hand-arm vibration	Breakers, grinders, needle guns (RA-17)	Credible exposure — unquantified	Tool selection, exposure-time management, health surveillance
Heat stress	All outdoor work (RA-16)	HIGH — Sindh summer; employer duties under the Sindh OSH framework [S-07]	Site heat-stress programme; measured-condition work/rest; acclimatisation; hydration/shade; heatstroke response
Ergonomics / manual handling	Blinds, valves, scaffold materials (RA-18)	Frequent moderate risk	Mechanical aids; task assessments; team lifts
Hydrocarbon / chemical spills	Refuelling, tie-ins, product use (RA-22, RA-19)	Credible — consequence-driven	Bunding, spill kits, drain protection, reporting and clean-up per environmental plan
Waste generation	Maintenance waste, contaminated absorbents/PPE (RA-22)	Routine — controlled	Segregation at source; authorised disposal route per current applicable Pakistan/Sindh requirements [CONFIRM]
Soil / drain contamination	Spills and washdown (RA-22)	Credible	Sealed work areas, no uncontrolled washdown, incident reporting
Water use and sanitation (hot Sindh site)	Welfare provision (Sec. 5/14)	Welfare-critical in heat	Potable water at workfronts, sanitation and rest facilities verified at mobilisation [CONFIRM provision]

14. Document limitations, assumptions and site-data request list

LIMITATIONS — BLUNT STATEMENT. This document is a publicly researched pre-assessment. It is based solely on the public sources listed in Section 17 (accessed 28 October 2026) and on established HSE practice for operating gas plants. The public record for Qadirpur is THIN and largely historical: detailed capacity/production figures are from 2004–2016, the Annual Report 2024 reproduction confirms Qadirpur only as one of OGDCL's major producing gas fields, and no current headcount, workfront, process-condition or equipment data is public. This document has NOT been verified by a site walkdown, operator or worker interview, approved drawings, current permits, process safety information, or a shift-specific headcount. It is not an official OGDCL document, not OGDCL-approved, not a contractor method statement, and not a substitute for the controlled plant HSE system, for task-level JSAs, or for specialist process studies (HAZID/HAZOP/LOPA/QRA). Controlled OGDCL HSE/PTW manuals were not available and were not reviewed [S-01]. Any person who treats this draft as authorization to work would be misusing it.

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ASSUMPTIONS MADE (all flagged for confirmation): (1) the selected boundary is the Qadirpur gas processing plant at Ghotki only; (2) the activity list reflects standard live-plant maintenance/modification interface tasks for a membrane gas-processing and compression plant of the type publicly described — each row applies only where the approved work pack confirms the activity, and the current plant configuration must be confirmed against current P&IDs because public process descriptions date from 2004–2017; (3) the proposed 5x5 matrix in Section 6.1 is used because the OGDCL matrix was unavailable; (4) permit and role names are generic functional descriptions pending mapping to OGDCL's controlled procedures.

14.1 Site-data request list

#	Site input required	Why it matters	Owner role	Priority
1	Approved site plot plan, exact assessment boundary, workfront and asset/unit IDs	Defines what this RA actually covers; without it every area reference stays [CONFIRM]	Operations / Document Control	Critical — before approval
2	Current work package: task list, method statements, JSA/JHA set and work schedule / turnaround scope if any	Confirms which register rows apply and which are N/A	Maintenance / Construction Manager	Critical — before approval
3	Current P&IDs, line list, equipment list; design pressure/temperature data	Required for isolation, line opening and pressure-test rows; no values are assumed in this draft	Process / Mechanical Engineer	Critical — before intrusive work
4	Current gas composition data incl. H2S and other hazardous constituents; SDS file for all products	Decides RA-03 and RA-19 controls; verification-critical hazards cannot be closed without it	Process Safety Engineer	Critical — before intrusive work
5	Hazardous-area classification drawings and electrical single-line drawings	Govern hot-work locations, temporary power and electrical work controls	E&I Engineer	Critical — before hot/electrical work
6	Underground services drawings and survey/scan records; ground-bearing/geotechnical data	Prevent service strikes and crane overturn (RA-12/RA-14)	Civil / Operations / Appointed Person	Critical — before excavation / first critical lift
7	Owner (OGDCL) risk matrix and risk-acceptance criteria	Replace the provisional PROPOSED 5x5 matrix and bands used in this draft	HSE Manager (owner)	Critical — before approval
8	Current OGDCL HSE manual and procedures: PTW, LOTO/isolation, hot work, confined space, line opening, excavation, lifting, pressure testing, radiography, commissioning, MOC, emergency plan	Public sources confirm OGDCL HSE guidelines exist and bind contractors [S-01]; the controlled texts were NOT available to, or reviewed for, this pre-assessment	Document Control (owner)	Critical — before approval
9	SIMOPS plan and live permit register access	Needed to operate RA-01 controls and daily permit coordination	Area Authority	Critical — before work starts
10	Actual headcount by shift: gate/access-control report, contractor mobilisation register, daily manpower report	Current total persons inside the boundary is not publicly disclosed; emergency headcount depends on it	Contractor Admin / Security	Critical — before work starts; then daily
11	Contractor/subcontractor register with scope, competence and authorisation records	Verifies who is on site and whether crews are authorised for their tasks	Maintenance / Construction Manager	High
12	Gas detection and fire-protection status at the interface; impairment/bypass register	Confirms credited barriers for hot work and line opening	Operations / Fire & Gas	Critical — before hot/intrusive work
13	Approved emergency response plan: alarm signals, muster points, routes, medical/ambulance provision, rescue capability, communications	RA-21 cannot be closed without it; no locations or response times are assumed in this draft	HSE Manager / Emergency Coordinator	Critical — before work starts
14	Project legal register for Pakistan/Sindh: Sindh OSH Act 2017 and Rules 2019 (official texts), petroleum/OGRA and environmental requirements applicable to the plant	This draft labels jurisdictions from secondary summaries only; exact current texts and applicability must be confirmed by the project's legal/HSE function — no clause numbers are cited from memory	HSE Manager / Legal	High — before approval
15	Site heat-stress procedure and measured-condition (e.g. WBGT) arrangements for the Sindh summer	Closes RA-16; no Pakistan midday-work ban is assumed	HSE Manager	High — before hot season
16	Site walkdown report, worker consultation records, incident/near-miss history and inspection findings authorised for this assessment	Grounds the RA in observed conditions instead of public sources alone	HSE Manager	High — before approval
17	Existing specialist studies: HAZID/HAZOP reports, LOPA/QRA where they exist, PSSR procedure	An activity RA does not replace these; interfaces must be confirmed (RA-02)	Process Safety Engineer	High
18	Welfare arrangements: drinking water, shade, rest areas, sanitation, clinic location [CONFIRM]	Supports heat-stress and welfare controls at a hot Sindh site	Maintenance / Construction Manager	Moderate

15. Approval, validation and review

15.1 Implementation, validation & review

Item	Requirement for this assessment
Critical controls / hold points	The critical controls in Section 8 and the hold point on each Section 7 register row must be verified with evidence before the associated work starts. No work within agreed protection zones of live equipment without signed operations authorisation.
Pre-start verification	PTW active, isolations proved (zero-energy verification), gas composition / H2S decision recorded (ACT-02), OGDCL matrix adopted (ACT-03), emergency arrangements briefed, toolbox talk completed and actual headcount capture running (ACT-06).
Review triggers	See Section 15.3 — any trigger event requires re-validation of the affected register rows before work continues.
Emergency and rescue arrangements	All operational details are critical site inputs from the approved site emergency plan (Section 12). Nothing in this pre-assessment states a muster point, response time or alarm level.

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15.2 Consultation & approval

Prepared by / role / date	Operations / supervisor review	HSE review	Authorized approver / date
Role: HSE — competent assessor Name: _____ Signature: _____ Date: _____	Role: Area Authority / Operations Representative (Maintenance/Construction Manager review also required — see note) Name: _____ Signature: _____ Date: _____	Role: HSE Manager Name: _____ Signature: _____ Date: _____	Role: as defined by the owner's document system Name: _____ Signature: _____ Date: _____

Additional review functions to be completed in the owner's document system: Reviewed by — Maintenance/Construction (Manager); Reviewed by — Process / Mechanical / Electrical discipline engineers (as applicable). Approval block intentionally left blank: names, signatures and dates may only be entered by the real authorized persons after site validation. Signing this Rev. 0 pre-assessment without closing the Critical site inputs in Section 14.1 would not constitute a valid approval for execution.

15.3 Review triggers

REVIEW TRIGGERS: change in scope, method, equipment or process status; new SIMOPS condition; any incident or significant near miss; change in gas composition or isolation boundary; new contractor or major workforce change; onset of the hot season or a severe-weather (monsoon) event; expired or failed critical control; expired permit conditions; or the review interval set by the owner's procedure — whichever occurs first.

16. Quality check

QA check	Result
All numeric figures carry a source / date / scope label	PASS — every figure in Sections 3 and 5 is tied to a source register entry [S-01..S-06] and labelled historical where it is historical
No company total presented as a Qadirpur figure; no headcount invented	PASS — OGDCL company totals are labelled company-only [S-06]; Section 5.2 is left as a site-record form; current headcount is stated NOT PUBLICLY DISCLOSED
No invented site count, workfront, process condition, chemical inventory, incident statistic, permit/drawing number, name, signature or approval	PASS — unknowns are [CONFIRM] / [SITE INPUT REQUIRED]; approval block blank; no current equipment list is asserted (public process descriptions are dated 2004–2017 and flagged)
Correct plant and jurisdiction; no cross-plant or cross-country confusion	PASS — single plant (Qadirpur, Ghotki, Sindh); other OGDCL fields and third-party power facilities are recorded as excluded; Pakistan/Sindh law is used and NO Saudi requirement (e.g. a midday-work ban) is imported
H2S treated correctly	PASS — RA-03 is VERIFICATION-CRITICAL: historical membrane-technology reporting [S-03] is cited as context only and current composition is stated NOT publicly confirmed; no concentration or alarm value is asserted
Every High/Extreme hazard has controls, a responsible role, verification evidence and a hold point / acceptance requirement	PASS — checked row by row in Section 7 and mirrored in the XLSX register
Risk scores internally consistent with the stated matrix; no residual reduction without a credible control rationale	PASS — scores recomputed programmatically; severity is not reduced for major-accident rows; the matrix is labelled PROPOSED
'Existing control' and 'additional action' clearly distinguished	PASS — separate columns; existing controls carry evidence status
Legal requirements vs technical guidance not misclassified	PASS — Sindh OSH Act/Rules labelled Pakistan/Sindh law (from a secondary summary, official text outstanding); ISO 45001 labelled a management-system standard; OSHA labelled US technical guidance only
Source links work; claim-level citations present; thinness of the public record disclosed	PASS — sources opened on 28 October 2026 are marked verified per source in Section 17; reproductions/secondary sources are flagged as such; the thin, historical nature of the record is stated on the cover, in Sections 2, 3 and 14
Tables legible; landscape used for wide registers; headers repeat; headers/footers and page numbers present	PASS — register, checklists, permit matrix and action tracker are landscape with repeated header rows; footer carries document ID, revision, NOT APPROVED FOR EXECUTION and page numbers
Document status states 'not approved for execution'	PASS — status banner on the cover and in every footer

17. References and traceability

Access date for all sources: 28 October 2026. Verification status is stated honestly for each source: reproductions and secondary compilations are flagged, not silently upgraded. Requirement types: [LAW] Pakistan/Sindh legal/regulatory, [OWNER] OGDCL-specific, [STANDARD] international standard, [GUIDANCE] technical guidance only.

Ref	Source (title — organisation, date)	Exact URL	Assertion(s) supported in this document	Verification status
S-01	Qadirpur Gas Field/Plant, Ghotki Sindh — Schedule of Requirement (contractor civil foundation work at Qadirpur Gas Plant) — Oil & Gas Development Company Limited (OGDCL) — official tender document, Tender TE/04/QP-PROCESS/2021 (document as published on ogdcl.com)	https://ogdcl.com/sites/default/files/tender-file/SOR-TOR-QP-P-04.pdf	Owner is OGDCL; the facility is named 'Qadirpur Gas Field/Plant, Ghotki Sindh'; contractors at the plant must follow OGDCL guidelines/policies related to HSE, and contractor scope there includes foundation/excavation, shifting and lifting work for stationary hydrocarbon plant equipment. Used ONLY to confirm owner, plant name/location and that contractor work of this type occurs under OGDCL HSE rules — NOT as a current equipment list or work pack.	Verified — opened 28 Oct 2026
S-02	Expanded OGDCL membrane-technology gas plant in service — Oil & Gas Journal (OGJ), 7 April 2004	https://www.ogj.com/refining-processing/article/17292506/expanded-ogdcl-membrane-technology-gas-plant-in-service	HISTORICAL CAPACITY ONLY: the membrane plant's 235 MMscfd capacity was expanded to 400 MMscfd of sales gas, with two further membrane skids expected to raise sales gas to 500 MMscfd by year-end 2004; pretreatment	Verified — opened 28 Oct 2026

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Ref	Source (title — organisation, date)	Exact URL	Assertion(s) supported in this document	Verification status
			included a chiller, a MemGuard unit and an improved Separex membrane system (UOP). These are 2004 figures — NOT current production or design data.	
S-03	Privatisation of Qadirpur gas field — Business Recorder, 25 October 2008	https://www.brecorder.com/news/amp/3565729	Field located about 8 km from Ghotki; discovered in 1990; joint-venture lease granted to OGDCL in October 1992 as owner/operator; regular production commenced September 1995 after installation of gas gathering and processing plant; developed in three phases until March 2004; the expanded membrane plant (2004) used UOP cellulose acetate plus technology for CO2 and H2S removal and hydrocarbon recovery; JV shares as reported in 2008: OGDCL 75%, PKP Exploration 9.5%, KUFPEC Pakistan 8.5%, PPL 7%. All figures are as reported in 2008 — historical context only.	Verified — opened 28 Oct 2026
S-04	Qadirpur gas field — Wikipedia (secondary compilation citing PPL/OGDCL and press sources), Page as accessed 28 Oct 2026 (content largely 2012-2017)	https://en.wikipedia.org/wiki/Qadirpur_gas_field	SECONDARY / HISTORICAL ONLY: location in Ghotki District, Sindh (8 km from Ghotki, 70 km NE of Sukkur, 100 km E of Jacobabad); discovered March 1990; production commenced September 1995; natural gas purified by a membrane separation process removing water vapour and carbon dioxide (UOP Separex); a 600 MMscfd gas processing facility was installed in 2006-07; compression facilities were being installed to address declining reservoir pressures; FY2015-16 average net production figures as stated on the page. No figure from this source is used as current plant data.	Verified — opened 28 Oct 2026 (secondary source; flagged accordingly)
S-05	Qadirpur Oil and Gas Field (Pakistan) — Global Energy Monitor (GEM.wiki), Published/updated 27 October 2026	https://www.gem.wiki/Qadirpur_Oil_and_Gas_Field_(Pakistan)	CURRENT STATUS: Qadirpur is listed as an OPERATING oil and gas field in Pakistan with Oil and Gas Development Company as operator. Production/reserves tables on the page are dated 2021 (and a 2008 design-capacity figure) and are NOT used as current rates in this assessment.	Verified — opened 28 Oct 2026
S-06	Annual Report 2024 — 'Shaping a Sustainable Future' (reproduction of the OGDCL report) — Oil & Gas Development Company Limited (via MoneyController reproduction), Report published 4 October 2024	https://www.moneycontroller.co.uk/finance-news/oil-gas-development-co-ltd/ogdcl-annual-report-2024-2445194	OGDCL's primary operations are exploration, development and production of oil and gas in Pakistan; Qadirpur is listed among OGDCL's major producing GAS fields (with Uch, KPD-TAY, Nashpa, Sinjhoro, Togh). Company-wide totals on the same pages (e.g. gas 717 MMcf/day) are OGDCL COMPANY totals — NOT Qadirpur figures — and are not used as plant data.	Verified — reproduction opened 28 Oct 2026; the original OGDCL PDF was not separately opened in this check (stated honestly)
S-07	Sindh Occupational Safety and Health Act, 2017 — summary of the Act and the Sindh OSH Rules, 2019 — HSE Forum (secondary summary of Sindh legislation), Summary published 2 October 2026; Act passed 17 Nov 2017, assented 26 Dec 2017, published as Sindh Act No. I of 2018	https://hseforum.pk/environment/sindh-occupational-safety-and-health-act-2017/	PAKISTAN / SINDH LAW (as summarised): the Sindh OSH Act, 2017 is the main provincial OSH framework in Sindh, applying throughout Sindh to places of work (excluding mines, subject to exemptions), with general duties of employers/persons responsible for workplaces, safety and health requirements, representatives/committees, and notification/investigation of accidents, dangerous occurrences and occupational illnesses; detailed requirements sit in the Sindh OSH Rules, 2019. Exact official texts must be confirmed in the project legal register — this summary is not the gazette text.	Verified — opened 28 Oct 2026 (secondary summary; official text is a site input)
S-08	ISO 45001:2018 — Occupational health and safety management systems — International Organization for Standardization (ISO), Published 2018; confirmed in 2024; Amendment 1:2024 (climate action) applies	https://www.iso.org/standard/63787.html	MANAGEMENT-SYSTEM STANDARD (framework only): ISO 45001:2018, confirmed in 2024, remains current with Amd 1:2024. It is NOT a site hazard checklist and NOT Pakistan law.	As verified in the master-prompt research dossier, 28 Oct 2026
S-09	Process Safety Management — Overview / Hazard Recognition — US OSHA, Current at dossier date	https://www.osha.gov/process-safety-management	TECHNICAL GUIDANCE ONLY (US regulator): process-safety concepts used to structure major-accident interface controls (RA-01/RA-02/RA-03). NOT Pakistan law and NOT a substitute for OGDCL process-safety requirements.	As cited in the master-prompt research dossier, 28 Oct 2026
S-10	Confined Spaces — Overview — US OSHA, Current at dossier date	https://www.osha.gov/confined-spaces	TECHNICAL GUIDANCE ONLY: confined-space control concepts supporting RA-06.	As cited in the master-prompt research dossier, 28 Oct 2026
S-11	Welding, Cutting and Brazing — Hazards and Solutions — US OSHA, Current at dossier date	https://www.osha.gov/welding-cutting-brazing/hazards-solutions/	TECHNICAL GUIDANCE ONLY: hot-work and welding-fume control concepts supporting RA-07/RA-08.	As cited in the master-prompt research dossier, 28 Oct 2026

Requirement-type key: the Sindh OSH Act 2017 / Rules 2019 [S-07] are Pakistan/Sindh LAW (official texts outstanding as a site input). OGDCL owner procedures (not reviewed) would be OWNER requirements [S-01]. ISO 45001 [S-08] is a STANDARD used as a management framework. OSHA material [S-09..S-11] is GUIDANCE from a US regulator and is never presented as Pakistan law. END OF DOCUMENT — Rev. 0 — Public-source pre-assessment.

Template use note: The completed assessment must reflect the actual work scope and current site conditions. Risk ratings, legal/owner references, workforce data, inspection observations and approvals must be verified by competent and authorized personnel. The generic reference format used for this document's layout does not establish compliance or owner acceptance.

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